

Verified Execution and Selective Memory for Financial Research Agents

Anonymous ACL submission

Draft under revision; not submission-ready.

Abstract

Financial research agents must retain relevant information, execute appropriate tools, and learn from outcomes that were actually available, yet success at one of these steps does not establish the validity of the final decision. We present FIN-SKILLS, a library for dated financial guidance, executable models and artifact-bound checks, and examine complementary mechanisms for selective memory: HiSTrim’s protected, sign-preserving context routing and an exploratory fruit-fly-inspired learner whose delayed feedback is checked before updating memory. The available financial evidence comprises a 12-run historical pilot, an audit feasibility study without independently gradable model submissions, and an autonomous-use development experiment in which two open-weight models produced 64 valid portfolio decisions and 43 successful tool calls, without improving mean proxy returns over no-library controls for either model. We separately retain the recommendation study’s reported HiSTrim comparisons and distinguish task accuracy, teacher agreement and resource use; these records motivate testing selective context within financial agents but do not establish a jointly evaluated system or a general economic benefit.

1 Introduction

A financial agent can call a forecasting package correctly yet use information that was unavailable when a trade would have occurred. It can select a familiar algorithm that does not fit the available data, or report performance from a backtest whose accounting assumptions differ from the stated trading rule. Access to more tools creates choices; it does not establish that those choices are appropriate or their outputs valid.

FIN-SKILLS addresses these connected problems through a reusable library for agents work-

ing with the Python quantitative-finance ecosystem. Its skills document library conventions, dated source checks, and failure modes. Executable interfaces let an agent inspect candidate methods, supply parameters, run supported algorithms, and receive structured results. Collection interfaces retain information timestamps, while research checks make missing evidence and detected defects explicit. The library packages existing methods rather than proposing a new forecasting or trading algorithm.

The agent remains responsible for selecting its tools. A deterministic recommendation service is available as one tool; it does not replace the model’s decision process. This distinction allows us to compare static guidance, predetermined tool outputs, and model-selected calls. A separate audit intervention binds check execution to the final submitted artifact, because stating that a check was performed is not evidence that it ran on the relevant code and data.

Selective memory raises a related question: what information should remain available to a decision maker? HiSTrim addresses this within a model’s context by retaining selected historical items while protecting task instructions. The fruit-fly-inspired component instead updates a persistent learned state from delayed outcomes. Neither operation validates a financial claim by itself. Our organizing principle is to keep information selection, execution evidence and feedback validity distinct, and to test their interaction rather than infer it from separate successes.

This working draft contributes a financial skill and model library, an artifact-bound audit protocol, and a formulation connecting protected context selection with checked learning feedback. The library supports direct human use and agent-selected execution; the two memory mechanisms operate at different stages and have different evidence. The HiSTrim recommendation study and

the financial-agent studies are evaluated separately. Their combination is a research design, not an already tested controller.

We ask whether final-artifact checks reduce incorrect accepted conclusions, whether autonomous tool access improves portfolio decisions, and whether checked delayed feedback can support adaptive learning. The current results support the feasibility of the LLM portfolio-decision loop. The memory application has implemented components and identified architectural limitations; its proposed persistence and exit policy requires further integration. No validated high-return strategy is claimed.

2 Related work

CheckList formalizes behavioral testing beyond aggregate task accuracy (Ribeiro et al., 2020). Financial backtest overfitting and selection bias pre-date LLM agents (Bailey and López de Prado, 2014; Bailey et al., 2017). Profit Mirage investigates information leakage in financial agents and proposes counterfactual interventions (Li et al., 2025). The Finance Agent Benchmark evaluates real financial research tasks (Bigeard et al., 2025). FinAgent combines multimodal financial information with trading tools, reflection, and memory (Zhang et al., 2024); StockBench evaluates sequential portfolio decisions using prices, fundamentals, and news (Chen et al., 2025). Thus tool-augmented trading, financial-agent leakage, and counterfactual testing are established research directions. Our focus is a reusable library and interventions that separate the effects of guidance, tool selection, and final-artifact checks. The short development study below is not a comparable leaderboard evaluation against these systems, and its results do not establish superiority.

3 Problem and framework overview

We distinguish three operations in a sequential research process. Context selection determines which available observations a model can attend to. Tool execution produces an analysis artifact and a record of the checks actually run on it. Learning uses a subsequent outcome to change persistent model state. A method can improve one operation while leaving the others unchanged or making the final decision worse.

Let $\mathcal{D}_{\leq t}$ contain records whose availability times are no later than decision time t . A context

selector produces C_t from those records and the task constraints. A fixed model chooses tools and an action from C_t , producing artifact A_t . An audit examines the relevant original inputs and A_t , not only the selected context. If a later outcome becomes available and passes the feedback checks, a learner may consume that outcome once. Retaining original records outside the model context allows an audit to inspect evidence that the selector omitted.

Two failure classes motivate the checks. Mechanical defects include unavailable inputs, future-survival filters and accounting mismatches. Interpretation defects arise when an agent treats a tool’s output, a popular source, or agreement with a reference model as proof of task validity. The source manuscript’s concrete failure examples motivate this distinction; their prevalence is not established by the current studies.

Table 1 specifies what each component changes. The implemented financial experiments exercise library access and audit enforcement. HiSTrim operates inside a backbone’s attention computation and requires access to its internal KV state; it cannot be assumed to work through an arbitrary hosted text-generation API. The persistent memory adapter has its own numerical backend. Connecting these interfaces requires a compatible execution backend and a new experiment.

4 Library design

Knowledge and executable interfaces. Domain skills route tasks to relevant libraries, while optional per-library skills describe conventions, dependencies, and failure modes in more detail. The Python package exposes catalog search, skill loading, and reference retrieval. Packaged skill documents and scripts are generated from the same repository sources. A verification date records a source check; it does not guarantee that a statement remains current. Table 2 describes the frozen study snapshot. These counts measure available interfaces, not independently validated capabilities.

Algorithm selection and execution. The algorithm catalog describes input requirements, optional dependencies, and supported tasks. Implemented families include portfolio construction, forecasting, volatility and risk estimation, technical signals, and execution schedules. Recommendation rules filter candidates by the request and

Component	Input and operation	Output to evaluate	Present evidence
HiSTrim context selection	Historical item representations; protect instructions and select retained KV entries	Task-label accuracy, teacher agreement, retained KV and measured latency	Recommendation manuscript results; financial integration untested
FIN-SKILLS execution	Available data, chosen algorithms and submitted artifacts; execute tools and required checks	Executed-call validity, audit completeness and independent research correctness	Financial regression checks and development studies
Checked persistent memory	Completed episodes with available, cost-inclusive outcomes; validate feedback before updating	Update validity, adaptation and net decision performance	Implemented interface; joint decision-policy evaluation pending

Table 1: Complementary operations and their evidence. The rows do not describe a single already deployed or jointly evaluated pipeline. KV context and learned persistent memory are different objects; none of the listed proxies alone establishes economic utility.

Component	Entries
Skill documents	129
Algorithm-backend catalog	48
Registered research guards	36
JSON-callable tools	56

Table 2: Inventory of the frozen study snapshot. Algorithm entries can expose existing methods through different backends. The autonomous study offers a price-compatible subset, not every catalog entry. This snapshot predates the September 22 RAG and Jev additions described below.

data requirements, then rank compatible entries using explicit preferences. They are not a learned predictor of future returns. A catalog entry does not imply that its dependency is installed or that execution will succeed; unavailable backends and execution failures remain explicit. Automatic execution does not silently substitute another method after a selected method fails.

Reusable models and lifecycle. The library supports direct Python use as well as agent-facing discovery and execution. Forecasting, regression, and classification adapters expose a common fit-predict lifecycle, including ARIMA, Ridge, random forests, LightGBM, and XGBoost when their dependencies are installed. Fitted models can be saved and restored with recorded environment and input metadata. This loader restores trusted user-created artifacts; it is not a pretrained-checkpoint download service. Portfolio optimizers and rule-based signals use execution interfaces without pretending to share a supervised training lifecycle. The current model registry also exposes an experimental fruit-fly memory adapter through `create_model("fly_memory")`. Its nu-

merical backend is the separately installed GPL-3.0-or-later `fin-skills-fly` extension and requires caller-supplied circuit parameters. It supports prediction, checked feedback updates and persistence; it does not implement a complete autonomous trading policy.

Built-in retrieval and model composition. The RAG interface chunks packaged skills or caller-supplied documents and retrieves passages using local BM25. Callers may supply embedding, reranking and text-generation functions. Passages retain their document identifiers, source offsets and availability timestamps; as-of retrieval excludes documents unavailable at the requested time. The index resides in memory and can be persisted as versioned JSON. Citation checks validate references to retrieved passage labels, not whether a passage supports a claim.

The shared model registry includes a hosted TypeSafe Jev adapter for structured decisions and passage reranking. It requires service access and explicit network permission; no Jev weights are distributed with the library. The adapter can serve as a RAG reranker while a caller-selected model generates the answer. These September 22 additions have interface tests, including mock Jev transports. Live-service inference, retrieval quality and reranking gains were not evaluated in the earlier studies. A paired development runner now freezes BM25 candidates, the time cutoff and context budget before Jev inference, with relevance scoring performed separately on saved outputs. Its public synthetic fixtures test this workflow. A quality comparison still requires independently annotated retrieval tasks and an unchanged answering model.

Information availability. News collection supports RSS/Atom feeds, static pages, GDELT, and disclosure sources. SQLite stores collection records, revisions, polling checkpoints and pending alerts; the operator explicitly starts collection. The package does not ship a populated financial dataset, and this record store is separate from the RAG index. Records distinguish publication time from observation time and retain revision information when available. An as-of digest filters future, stale, and undated records and deduplicates canonical URLs. These mechanisms support timestamp-aware processing; they do not reconstruct an independently validated historical news archive. Keyword risk indicators are not calibrated sentiment estimates, and absent news is treated as unknown rather than evidence of low risk. News collection is not exercised in the portfolio experiment reported here.

Model choice and rule-based advice. An agent can read a skill, inspect available algorithms, execute a selected method with its own parameters, and use the returned result in its next decision. The library also provides an optional market-state adviser based on trailing price and volatility statistics. Its trend, range, and stress categories are rule outputs, not learned regime probabilities or validated predictions of which strategy will outperform. A suggested strategy is therefore additional evidence the model may inspect. The autonomous study records the model’s actual calls; it does not equate access to this adviser with autonomous algorithm selection.

Research workflow and boundaries. Research interfaces support data preflight, candidate evaluation, chronological validation, holdout reporting, and trial records. Holding final data out of candidate ranking reduces one avenue for selection bias, but the caller must still provide valid timestamps and data provenance. Python and JSON tool interfaces expose structured results; guards requiring live Python objects are not all available as JSON calls. Neither a data record nor a successful tool call certifies the whole research process.

5 Selective context with HiSTrim

HiSTrim contributes a concrete mechanism for selecting historical context while retaining task constraints. The source study uses chronological interaction items for recommendation. A financial

application would instead need to define meaningful record spans and preserve timestamps, asset identifiers and task requirements. This mapping is a proposed application; it is not part of the completed financial runs.

5.1 Protected, sign-preserving selection

Let $\mathcal{H}_{\ell-1}$ be the historical items surviving before pruning layer ℓ , and let $\mathcal{P}$ be a disjoint set of protected token positions. For an item representation h_m , the dual router computes

$$g_m = w_g^\top h_m + b_g, \quad u_m = w_u^\top h_m + b_u, \quad (1)$$

$$r_m = \sigma(g_m)u_m, \quad \sigma(g) = (1 + e^{-g})^{-1}. \quad (2)$$

For finite real logits, $\sigma(g_m) > 0$, so $\text{sign}(r_m) = \text{sign}(u_m)$; holding g_m fixed gives $\partial r_m / \partial u_m = \sigma(g_m) > 0$. By contrast, $\text{SiLU}(g_m)u_m$ is positive when both logits are negative. The sigmoid form removes this algebraic sign reversal. It does not prove that the utility score is correct or that thresholding will retain every useful item.

The router standardizes scores within a sequence and applies a learned threshold:

$$z_m = (r_m - \mu_\ell) / (s_\ell + \epsilon), \quad (3)$$

$$\mathcal{H}_\ell = \{m \in \mathcal{H}_{\ell-1} : z_m \geq \tau_\ell\}, \quad (4)$$

$$\mathcal{T}_\ell = \mathcal{P} \cup \bigcup_{m \in \mathcal{H}_\ell} \mathcal{I}_m. \quad (5)$$

Here $\mathcal{I}_m$ contains item m ’s token positions. This separates item indices from token indices and ensures $\mathcal{H}_\ell \subseteq \mathcal{H}_{\ell-1}$. Protected positions remain in $\mathcal{T}_\ell$. Their retention prevents literal deletion, but changes in surrounding history can still change their hidden states and the model’s answer. No equality of mutual information follows from masking alone.

5.2 Router fitting and physical compaction

The source method initializes routers from ridge-regression directions fitted on calibration data, normalizes each direction, and normalizes their weighted average. For nonzero fitted directions v_s , nonnegative weights summing to one, and a nonzero average, this gives

$$w_0 = \frac{\sum_s \alpha_s v_s / \|v_s\|_2}{\|\sum_s \alpha_s v_s / \|v_s\|_2\|_2}. \quad (6)$$

This construction fixes the norm; variance reduction needs additional distributional assumptions.

A zero direction or cancelling average requires explicit handling. Forward-only refinement evaluates paired perturbations and updates directions in the tangent space, using $g_{\perp} = (I - ww^{\top})g$ for unit w , followed by normalization. Thresholds are updated separately. These operations describe an optimization procedure, not a guarantee of convergence or avoidance of threshold collapse. Fixed numerical stabilizers also limit exact scale invariance of standardized scores.

At inference, HiSTrim-VarLen gathers retained entries into packed buffers with sequence offsets. Physical gathering distinguishes a smaller cache from a mask that leaves all entries allocated. Real memory and latency depend on the backend, sequence lengths, beam sharing, packing overhead and retained fraction; they must be measured for the model executing the financial task.

5.3 Attention compensation and its limits

Eviction changes attention normalization. For a fixed query, let s_0 be a sink logit and let $\mathcal{E}$ contain the removed positions, excluding the sink. Replacing its exponential mass by the sink plus the removed mass would require

$$\Delta^* = \log \left(1 + \sum_{j \in \mathcal{E}} e^{s_j - s_0} \right). \quad (7)$$

Then $e^{s_0 + \Delta^*} = e^{s_0} + \sum_{j \in \mathcal{E}} e^{s_j}$ exactly for these fixed logits. Even this identity preserves only the denominator: preserving the attention output also requires the corresponding value-weighted contribution.

The source’s practical offset $\beta \log(|\mathcal{E}| + 1)$ approximates the missing mass using only a count. Equality with Equation 7 requires a specific relation between removed logits and the sink; it is not an unconditional conservation theorem. We therefore retain it as an attention-compensation heuristic whose effect must be measured, together with protected spans and sign-preserving routing. These qualifications preserve the method without attributing guarantees that the stated assumptions do not establish.

Evidence and transfer. The recommendation study separates semantic-ID generation, rating prediction and hybrid ranking. Appendix A preserves its reported results and their provenance. Teacher agreement and label accuracy answer different questions; neither directly measures finan-

cial research correctness. A joint test must compare the same financial model with and without context selection under matched information and audit access, rather than transfer a recommendation gain to a return claim.

6 Audit contract

Python guards inspect supplied artifacts and return structured findings. The following receipt mechanism belongs to the experimental audit controller; it is not an automatic property of every library call.

Audit completeness. A task specifies a nonempty set of required guards. A report is `PASS` only when all required checks ran and passed with no rejected input; a detected defect yields `FAIL`, while missing or rejected checks yield `INCOMPLETE` in the absence of a detected failure. Warnings remain visible. A passing subset is not a certificate that the whole research task is valid.

Execution evidence. The controller runs checks on a snapshot of the actual submission and supplied data. It records each result, elapsed time, and a SHA-256 digest of the code, input files, manifest and numerical report claims. Changing those artifacts invalidates earlier receipts. Citation-list changes alone do not invalidate a check. Unknown guard names and execution errors cannot create successful execution records. Executed-but-failed, unexecuted, and stale checks are distinct states.

The current intervention binds the library’s close-price causality and survivorship guards to submitted code and positions. It additionally checks reported accounting and samples same-session dependencies. These checks cover a subset of possible defects. The runner exposes a fixed guidance excerpt, not the entire skill collection.

7 Independent evaluation

A seeded synthetic market provides quoted prices, corporate actions, listings, post-close commentary, precomputed scores and fundamental disclosures. A submitted function returns portfolio weights; a separate report states its net Sharpe ratio. The task now specifies that session- t weights must use information available by the end of session $t - 1$, consistent with close-to-close return accounting.

The independent evaluator redraws future data and individual sessions, measures changes in

Historical arm	Mean absolute Sharpe gap
Opus, no library	0.188
Opus, library	0.034
Haiku, no library	0.574
Haiku, library	11.949

Table 3: Exploratory pilot: three runs per arm. Values come from saved grades; model versions and full interaction-level execution provenance require further verification.

submitted positions, checks post-delisting exposure, and recomputes performance at a fixed transaction-cost convention. Public intervention checks do not call this oracle. Private grades are generated only after the submission receipt is saved and are not returned to the agent for repair. Rejected submissions are also graded when executable.

The oracle’s fixed-cost Sharpe and the public tool’s stated-cost Sharpe answer different questions. Their discrepancy can reflect accounting conventions as well as coding errors. Neither should be called an estimate of genuine alpha without timing and other validity checks. A latent-drift strategy’s sample Sharpe is a reference observation, not a theoretical upper bound. Exceeding it is not mathematical proof of leakage.

8 Evidence available so far

8.1 Synthetic regression checks

Saved robustness results record 36 detections among 36 planted instances: twelve defect families repeated across three seeded worlds, with no reported clean-control false alarms in those worlds. This is regression evidence within one synthetic design. Repeating a seed design does not supply independently authored defects or estimate a general real-world false-positive rate. The detection suite exercises thirteen guards, not every registered guard.

8.2 Historical agent pilot

The saved pilot comprises three markets, two model tiers, and library-present versus library-absent conditions, with one run per cell. Table 3 recomputes the mean absolute discrepancy from the saved raw grades. The observed relative reduction for Opus is 81.9%, computed before rounding. It is not a significance claim.

The Haiku library arm has the largest discrepancy in this pilot. This motivates testing whether

guidance, tool use, or other differences explain the outcome. It does not show that documentation causes weaker models to fail or that mandatory checks solve the problem. Historical reports of unexecuted guard citations require original execution logs for verification; the saved grade table alone does not establish that behavior.

8.3 Numerical and real-data checks

The numerical benchmark compares implementations with reference calculations using case-specific tolerances. Passing within tolerance is not exact equality. Runtime measurements report the names and number of guards actually exercised, skipped and rejected checks, and machine-dependent timing. A fitted slope across a few data sizes is descriptive, not an asymptotic complexity bound.

The KOL fixture supports recomputing profile-group aggregates. Its model prediction metrics are saved summaries, not values independently recomputed from predictions and outcomes. In particular, its stored pooled Rank IC and mean daily cross-sectional Rank IC can have different signs. We do not treat this fixture as end-to-end reproduction, evidence of profitable trading, or external validation of the agent intervention. A separate evaluator now requires timestamped row-level predictions and outcomes, reports both forms of IC, and uses paired date-block uncertainty. Those source records remain an outstanding evidence requirement.

9 Four-condition study

The new protocol compares task guidance with ordinary accounting feedback, the same setup plus a fixed skill excerpt, the excerpt plus optional executable guards, and the excerpt plus a mandatory final audit. Market data, inference seeds and budgets are paired. Every condition has the same turn and response-token caps; actual token use, execution time and tool calls are also recorded. Conditions run in a frozen randomized order. Failed requests have no hidden retry or synthetic response fallback.

The primary outcome is incorrect accepted submissions divided by all attempted submissions. Correctness among accepted outputs and overall acceptance rate must be reported alongside it, so refusing every output cannot appear to solve the task. Secondary outcomes include independent

leakage measures, reporting discrepancy, post-delisting exposure, unexecuted and stale citations, repair success and resource cost. Missing, rejected, ungradable and provider-failed runs remain separate outcomes.

The first Beacon run is a public-seed feasibility study using pinned open-weight models. It tests operation and supports cost and sample-size planning. A confirmatory study additionally needs independent defect authorship, retained clean cases, repeated runs, and verified scorer isolation. Market-level paired uncertainty must preserve repetitions within markets. The present protocol is an amendment after examining the pilot, not a preregistration of its observations. Failure to find a return difference would not establish equivalence.

Operational findings. The first Beacon pre-flight rejected valid tool-call JSON wrapped in Markdown fences. We archived those records and amended the parser before a separate batch; this was an interface failure, not evidence of financial reasoning failure. In the revised batch, Qwen2.5-Coder 7B and 14B each completed 12 recorded cells, with 0 and 7 accepted submissions respectively. Neither model produced an independent numerical grade: final code failed input or output validation, or a required artifact was absent. Accepted does not mean correct, and these observations cannot estimate improvement in financial integrity.

Table 4 shows the per-condition breakdown for 14B, the only model that produced accepted submissions. Acceptance rate was 67% under C0 and C1, 100% under C2 (optional guards), and 0% under C3 (enforced audit): all three C3 runs reached the turn limit without producing a passing submission. Mean token use under C3 was 47,491 versus 13,877 under C2, indicating substantially greater iteration before the turn limit was hit. All accepted submissions were ungradable; the primary outcome—incorrect accepted submissions per attempt—cannot be estimated from this batch.

The 32B run encountered nonfinite generation values and a CUDA error before a tool call. Subsequent failure records from that process are not independent capability observations. At the time of that feasibility report, a separately recorded single-GPU recovery run was queued. A handwritten causal control passed the mandatory audit and in-

dependent timing/accounting checks on one public market. That control verifies an executable path, not general scorer validity. Before a confirmatory study, basic artifact validity must be distinguished from the incremental contribution of financial checks.

10 Autonomous library-use development study

The preceding audit study concerns submitted research artifacts. This separate experiment asks whether library access improves portfolio decisions made by a fixed model. An initial 12-trial pilot supplied predetermined tool results; it did not test autonomous selection. That distinction motivated the corrected intervention below. Both experiments use the same public market path, so the second is explicitly development evidence.

10.1 Matched controls and model-selected tools

We compare Qwen2.5-Coder 7B and 14B under four conditions: historical prices and basic summaries alone; the same inputs with fixed skill excerpts; excerpts plus predetermined tool outputs; and autonomous access to library tools. Each condition uses inference seeds 11 and 23 and the same maximum of five model responses per decision, each capped at 256 generated tokens. Conditions run in a frozen randomized order. Identical limits do not imply equal actual consumption, because the agent may finalize early.

The autonomous condition starts with tool descriptions. It can discover the algorithm catalog, read a related skill, run an algorithm with chosen parameters, or ask an optional heuristic strategy adviser about an asset. The model chooses whether to call a tool, which tool and parameters to use, and whether to make another call after inspecting the result. No particular call is forced. The dispatcher executes the requested API against the current past-only snapshot and returns its actual output. Tool requests and results are recorded separately from model explanations; saying that a tool ran does not count as execution. The offered subset contains 18 price-compatible algorithms, not every capability in the library.

All conditions can finalize or use a response for further assessment. Invalid requests and formatting errors consume the existing budget and return feedback. A decision without a valid final allo-

Model	Condition	Accepted/Planned	Mean tokens	Primary status
14B	C0 no library	2/3	13,535	1× turn limit
	C1 text only	2/3	13,095	1× turn limit
	C2 optional guards	3/3	13,877	all accepted
	C3 enforced audit	0/3	47,491	3× turn limit
7B	all (12 cells)	0/12	—	turn limit / error
32B	all (12 cells)	0/12	0	provider error

Table 4: Per-condition Beacon feasibility results. All accepted 14B submissions were ungradable; mean tokens are averaged over all completed cells in the condition. 32B token count is zero because the process failed before any tool call. Values come from `BEACON_FEASIBILITY_RESULTS.json`; numbers in `evidence_numbers.tex` are generated by `scripts/build_paper_evidence.py`.

cation leaves existing holdings unchanged and remains a failure in the record; provider errors abort a run. Responses are saved before dispatch, and final allocations are frozen before independent scoring. This is a JSON action loop, not provider-native function calling or arbitrary generated-code execution.

10.2 Data, timing and scope

The existing ECB fixture supplies USD, JPY, GBP and CHF reference rates quoted per EUR. Their reciprocals represent foreign-currency values in EUR. These are indicative reference prices, not executable quotes; the ledger excludes interest and carry. Model messages anonymize asset identifiers and omit dates, which does not establish absence of pretraining contamination. No historical news archive is included.

Each decision sees 126 past observations. Evaluation covers the first 84 reference observations of 2025, ending May 2, with four decisions spaced 21 observations apart. Decisions after observation t execute at the reference price at $t + 1$ and cannot earn the preceding movement. Allocations are long-only, without leverage; residual weight remains zero-interest EUR cash. The primary cost is 5 basis points per side, including entry and terminal liquidation. Zero- and 20-basis-point scenarios rescore the same frozen decisions. Cash, equal-weight buy-and-hold and deterministic library rules are separate controls. Two inference seeds on one path are not two independent markets.

10.3 Completed runs and outcomes

The preliminary fixed-output pilot completed 12 trials and 48 valid decisions. At 5 basis points, mean full-context minus no-library proxy return was -0.455 percentage points for 7B and $+0.449$

points for 14B. This mixed result cannot answer the autonomous selection question.

The autonomous-loop implementation initially rejected single Markdown fences around valid JSON, although the earlier final-action parser accepted them. We retained and cancelled that instrument-failure attempt, then corrected the parser consistently for all action types. The corrected run used unchanged prompts, models, seeds, tools and market window in a new workspace. Its instrument suite passed 26 tests, including past-only tool inputs, optional calls, error recovery, turn limits and fenced responses.

The corrected experiment completed all 16 trials and 64 final decisions. Autonomous 7B made 13 successful tool requests and autonomous 14B made 30, with no call or format errors in these conditions. Across all conditions, one unavailable-tool request was denied and three format errors occurred in the 7B text-only condition; the model recovered within its existing budget. Valid final decisions therefore do not imply error-free inter-action traces.

The autonomous agents used different tools: 7B executed cross-sectional momentum, Bollinger reversion and volatility-targeted momentum, while 14B mostly executed Bollinger reversion and also read the trend-following skill. These observed choices establish that the model selected calls. They do not establish that selection was appropriate or optimal.

Table 5 reports every condition. Autonomous access underperformed the no-library condition by 4.084 percentage points for 7B and 0.459 points for 14B; it also underperformed fixed tool outputs by 0.909 and 0.753 points. Autonomous 14B outperformed its text-only control by 0.783 points. Thus successful tool execution did not translate into a mean advantage on either primary compari-

Condition	7B	14B
No library	+0.100	-3.324
Skill text	-2.930	-4.567
Fixed tool outputs	-3.075	-3.030
Autonomous library	-3.984	-3.783

Table 5: Mean cumulative proxy return (%) across two inference seeds at 5 basis points. These are indicative FX price-only results, not executable investment returns. The window was inspected during the earlier pilot; this is development evidence.

son in this development run.

Risk exposure differs materially: mean risky exposure for 7B was 0.500 without the library and 0.969 with autonomous access; for 14B it was 1.000 and 0.500, respectively. The results are whole-policy comparisons, not risk-matched alpha estimates. Autonomous groups used 38 model calls per model across eight decisions, compared with 8 and 12 calls for the 7B and 14B no-library groups. The complete run consumed 156 calls, 1,155,782 prompt tokens and 10,165 completion tokens. Economic outcomes must be interpreted alongside this additional computation.

The same prompt, market window and seeds were retained after the parser correction; no unfavorable financial run was removed. Paired date-block intervals are descriptive for this short path, not confirmatory tests. A paper-level claim requires untouched evaluation periods, independent markets, executable-price conventions, risk-matched controls and a competing tool-access baseline. This experiment demonstrates autonomous use and independent scoring, but not a general return benefit from the library.

11 Fruit-fly-inspired learning with checked feedback

This application extends the library from assisting fixed models to checking the experience used by an online learner. Its research question is whether memory operating over different timescales, updated from checked delayed outcomes, can support decisions to enter, persist, exit, and revisit opportunities under changing conditions and transaction costs. These policy capabilities remain development objectives.

Memory component. Our biological starting point is the mushroom-body memory model of Huang et al. (2024), which studies interactions between short- and long-term memory units. The

repository separates a numerical reproduction harness from an online adaptation. The adaptation advances its retention clock using observed conditioning events and reads memory without changing its state. Mapping signed financial returns to reinforcement and interpreting neuronal readouts as preferences are engineering choices. They are not validated biological correspondences, and the readout is not a calibrated expected return. The reproduction and derived model code are distributed as a separately installed GPL-3.0-or-later extension. The core library provides its adapter and model-card interface without bundling the extension or circuit parameters.

Connection to the library. The implemented feedback interface treats an entry and its subsequent holding periods as one completed decision episode. Given positive, cost-inclusive account values $W_0, \dots, W_T$, its learning target is

$$G = \sum_{t=0}^{T-1} \log(W_{t+1}/W_t) = \log(W_T/W_0). \quad (8)$$

This undiscounted complete-episode target includes entry costs and later holding outcomes. It requires the episode to end before the result can be used. The interface checks interval continuity, return accounting, and availability timestamps, and invokes the library’s source-lineage guard. Pending feedback waits; invalid feedback is rejected; valid negative returns remain learning signals. A completed receipt can be consumed once. The source account values still require independent execution reconciliation: these checks cannot establish that supplied values are authentic.

Current evidence and remaining comparison.

Earlier custom sparse controllers made market decisions, but their development results did not establish a consistent financial advantage. Architecture diagnostics identified a mismatch between one-interval rewards and persistent holdings, return clipping that can change preference ordering, and impaired adaptation under repeated reward reversals. Those findings concern the tested implementations, not the suitability of mushroom-body models in general. The newer memory interface uses supplied outcomes in controlled learning cases; it does not yet autonomously choose entry, holding duration, or exit. It is separate from the LLM tool-use experiment in the preceding section.

The compact-memory benchmark now includes a frozen-memory control with the same volatility gate as the learned-memory arm, using identical data, costs, chronological splits and seeds. This separates the contribution of learning from that of the risk gate. No result for the newly added control has been recorded. The volatility gate controls exposure; it is distinct from the delayed-feedback checks described above.

The proposed comparison will hold information, actions, costs, and learning targets constant across controllers with and without feedback checks. Linear and dense controllers, non-biological models with matched memory timescales, and ablations of memory interactions are needed to distinguish the contributions of feedback validation and memory structure. Evaluation must measure adaptation, exit behavior, turnover, and net performance on untouched periods. This model-development contribution is therefore an implemented research direction with explicit tests still required, not evidence that biological inspiration or library access yields higher returns.

12 Discussion and conclusion

The studies expose different gaps between a plausible decision and a supported result. The financial-agent loop demonstrates that models can choose and execute tools, but successful execution did not improve the primary proxy-return comparison. Audit receipts make the scope of executed checks explicit, while the feasibility study shows that artifact validity must be established before financial correctness can be measured. The recommendation study supplies a concrete context-selection mechanism and reported task-specific comparisons, with its evidence status recorded in Appendix A. Checked episode feedback addresses a further issue: whether a learning update uses an outcome available at the time it is consumed.

A joint evaluation should cross context selection with audit access while holding the backbone, information cutoff, action space and compute budget fixed. Feedback validation requires a separate matched learning ablation. Such a design can distinguish context savings, correct execution and adaptation from changes in risk exposure or available information. The present contribution is a library and a testable account of these interfaces, supported by separate development studies rather than a claim that the complete combination im-

proves financial returns.

Limitations and remaining work

The Beacon worker uses Linux Landlock and seccomp to restrict filesystem access, network calls and process creation; unsupported enforcement fails closed. Local subprocess execution alone provides no such isolation. Submitted Python still shares an interpreter with individual checks, so host confinement does not establish that the evaluator resists tampering. A formal adversarial or private-holdout claim needs an independently protected scorer and additional validation. Publicly inspected generator code and seeds are development data. Finite perturbations can miss conditional dependencies; stochastic strategies need repeatability controls. A passing audit is limited to its input assumptions and required checks. The full multi-model study and row-level real-world reproduction are not yet established. The autonomous-use development experiment adds a functioning portfolio-decision loop; it does not resolve those limitations or establish profitable trading. Its reference-price data, limited market coverage, unequal actual compute use and previously inspected window preclude a confirmatory economic-utility claim. HiSTrim’s underlying recommendation run records were not available for this merge; its retained results are explicitly author-reported. Context compaction can discard necessary evidence even when protected tokens survive, and the proposed connection to financial agents has not been evaluated. Resource savings alone do not establish lower energy use or a better research decision.

Authorship, data and disclosure status

Author order, affiliations, funding, data permissions and final declarations require confirmation by the accountable authors. The earlier draft’s unconfirmed organization and email details have been removed. AI assistance was used for code review, revisions and experiment preparation in this work session; the final disclosure and human review record must be completed before submission. The manuscript remains a working draft.

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A Recommendation evidence retained from the source manuscript

This section preserves the substantive HiSTrim comparison from the shared manuscript. The source is the Overleaf revision uploaded by Shwai He on September 22, 2026, with the interface displaying 19:05. Its TeX and figures were downloaded before merging. The tables below are author-reported manuscript values, not an independent replication: the original training code, user-level predictions and run receipts were not available in this workspace. The local merge record stores source hashes and arithmetic checks. This evidence status is separate from the financial result files used elsewhere.

Generative recommendation and baselines.

Table 6 retains all five task columns and all six method rows per model scale from the source’s principal generative table. Keeping every row shows where other methods outperform HiSTrim, including the Video task. The table does not support universal superiority across tasks. Reported KV retention also differs between methods, so these rows are not perfectly matched resource budgets.

Calibration-seed variability. The source appendix lists five calibration partitions for OneRec-1.7B Ad. Recomputing from its rounded rows yields Pass@32 of $18.78\% \pm 0.72\%$ and MRR@32 multiplied by 100 of 3.81 ± 0.19 (mean and sample standard deviation). The Pass@32 mean resolves the source main text’s differing rounded value. These are variations across fitted routers evaluated on the stated test set, not independent test populations. Significance claims require the original paired predictions and sampling procedure and are not inferred from these summary rows.

Task accuracy, agreement and cost. Table 7 retains both favorable comparisons and a counterexample to treating teacher agreement as task success. On Beauty, the source reports lower KL for HiSTrim than for TokenDrop-50, while its label MAE is higher. On the hybrid ranking task, the aggressive variant is faster but has lower NDCG than the dense baseline. The Goodreads comparison reports lower MAE with reduced KV; the original prediction records remain necessary to assess uncertainty. These contrasts motivate joint measurement of quality and resources rather than a single aggregate claim of improvement.

Items requiring reconciliation. The source’s figure captions and tables contain different aggregate configurations; we retain the explicit table rows rather than combining their maxima. The source also describes several calibration-set sizes without a complete run-to-configuration map. Hardware timings, significance markers, broad baseline compatibility claims and the proposed transfer to financial agents require their supporting run records. The untouched source remains available in the pre-merge backup. These gaps do not alter the financial study’s recorded outcomes and are not filled with new experiments or assumed results in this merge.

Model	Method	Hist. KV (%)	Ad	Product	Video	Label-cond.	Interactive
1.7B	Full Dense	100.0	3.24	7.81	5.16	1.25	11.95
1.7B	TokenDrop-50	50.0	3.08	7.72	4.89	1.29	11.70
1.7B	H2O	54.6	3.98	7.51	7.77	1.25	11.72
1.7B	SnapKV	54.6	3.85	7.75	7.36	1.30	11.85
1.7B	Router-Tuning	45.7	4.00	7.27	3.30	1.18	11.24
1.7B	HiSTrim	47.5	4.10	8.09	6.43	1.50	12.08
8B	Full Dense	100.0	4.56	8.22	7.67	1.41	18.25
8B	TokenDrop-50	50.0	4.22	7.89	7.12	1.32	17.40
8B	H2O	54.6	5.06	8.28	8.94	1.37	18.95
8B	SnapKV	54.6	5.02	8.25	8.54	1.72	18.70
8B	Router-Tuning	46.2	4.12	7.95	6.84	1.34	17.65
8B	HiSTrim	49.5	5.25	8.54	7.96	1.47	19.21

Table 6: Author-reported generative recommendation results retained from the Overleaf source table: MRR@32 multiplied by 100, with the source reporting 1,000 users per task and 32 beams. Historical KV retention excludes protected tokens. These manuscript values have not been reproduced from predictions; source significance markers are omitted pending the underlying paired records.

Task / model	Method	Hist. KV (%)	Label metric	Teacher metric	Speedup
Beauty / Qwen3-0.6B	Dense	100.0	MAE 1.642	KL 0.000	—
	TokenDrop-50	50.0	MAE 1.517	KL 0.233	—
	HiSTrim	50.1	MAE 1.643	KL 0.084	—
Goodreads / Qwen3-8B	Dense	100.0	MAE 1.232	—	—
	HiSTrim	35.0	MAE 1.078	Teacher MAE 0.651	—
Amazon / DCNv2+LM	Dense	100.0	NDCG@10 0.5183	—	1.00×
	TokenDrop-50	50.0	NDCG@10 0.5204	—	1.79×
	HiSTrim	50.0	NDCG@10 0.5567	—	1.78×
	HiSTrim aggressive	35.0	NDCG@10 0.5065	—	2.82×

Table 7: Selected author-reported rating and ranking comparisons from the source table. The source reports 357 Beauty, 250 Goodreads and 40 Amazon ranking cases. Lower MAE and KL are better; higher NDCG is better. Missing entries are not measured zeros. Teacher agreement, task-label performance and speed are kept in distinct columns.